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POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
QUEZON CITY CAMPUS

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements is to describe EventSync in a clear and testable manner. This explains what the system is supposed to do, who will use it, which constraints will apply, and how the system is going to behave under normal use. The document is intended for the project stakeholders, instructors, and developers who need a shared 13 reference for the proposed system.

1.2 Scope

EventSync is a web-based Event Organizer Management System designed for school environments. It focuses on event requests, schedules, venue reservations, notifications, and dashboard monitoring so that the student organizations, faculty members, administrators, and event double bookings.

1.3 Glossary

Term	Definition
Administrator	User with full control over approvals, records, and system settings.
Organizer	User who creates events, submits reservations, and manages event details.
Participant/Attendee	User who views event details and receives notifications.
Event	A scheduled school activity such as a seminar, meeting, workshop, or program.
Venue	A physical room or space that can be reserved for an event.



Reservation	A request or booking for using a venue on a specific date and time.
Notification	A reminder or alert issued by the system.
SRS	Software Requirements Specification.
UI	User Interface.
DB	Database.

1.4 *References*

IEEE Recommended Practice for Software Requirements Specifications (IEEE 830 style structure).

1.5 *Overview*

Section 2 gives the overall description of the system, including the product perspective, user classes, operating system, and constraints. Section 3 presents the detailed requirements, including external interfaces, functional behavior, non-functional behavior requirements. Section 4 summarizes other supporting information.

2. Overall Description

2.1 *Product Perspective*

EventSync is an independent web application that goes between the end users and the school event data it manages. Users access the system through the use of web browsers. The web application is connected to a backend service that handles authentication, reservations, scheduling, notifications, and dashboard information. The backend stores records in a MySQL database and uses a scheduler for timed notification reminders.

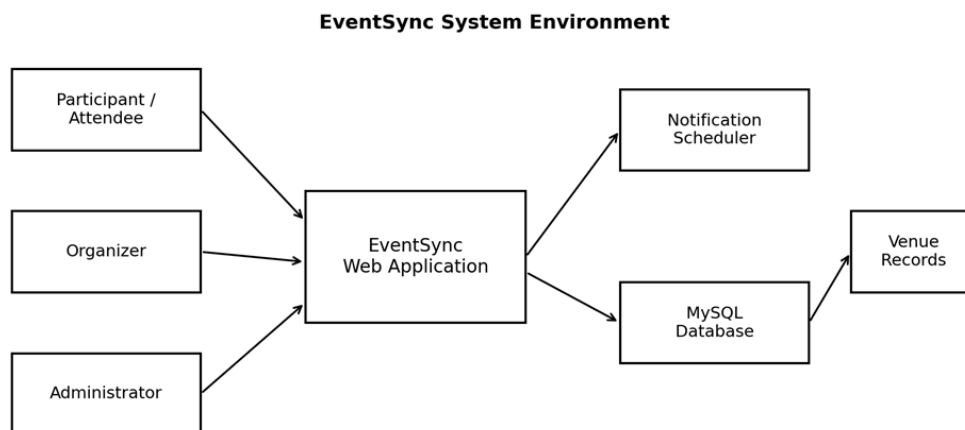


Figure 1. EventSync system environment

2.2 *Product Functions*

- User registration, login, and role-based access control.
- Creation, updating, approval, and tracking of events.
- Venue browsing, availability checking, and reservation management.
- Automated notifications and reminders for event-related updates.
- A centralized dashboard for monitoring active events and reservations.

2.3 *User Characteristics*

User Class	Description	Typical Needs
Administrator	Manages user access, approves or rejects reservations, and monitors overall activity.	Full access, clear dashboards, fast approval tools.



Organizer	Creates events and submits venue reservations for approval.	Simple forms, venue availability, status updates.
Participant / Attendee	Views event details and receives reminders.	Easy access to schedules and notifications.

2.4 *Operating Environment*

- Modern web browser on desktop or mobile devices.
- Server-side application built with Node.js and Express.js.
- MySQL database for persistent data storage.
- Notification scheduler for automated reminders.
- Stable internet connection for normal operations.

2.5 *Design and Implementation Constraints*

- The system is intended for school-based event management
- Venue availability must overlook and prevent conflicting bookings.
- Access to sensitive actions must be limited by user role.
- The system must support near real-time updates for schedules and reservations.
- The user interface must remain responsive and easy to understand.

2.6 *Assumption and Dependencies*

- Users will have valid internet access and an active browser.
- School venues and event details will be maintained by authorized personnel.
- The system depends on the availability of the MySQL database and backend service.
- Notification timing depends on the server scheduler and system clock.



3. Requirements Specification

3.1 *External Interface Requirements*

The system has a modular design, it contains a clean and consistent layout across all the pages. It also allows the forms, tables, calendars, and alerts in a readable format. The system ensures accessibility between desktops and mobile devices. The software interface relies on a stable backend infrastructure where the system communicates directly with a MySQL database for all record storage needs. The application functionalizes data management by using server-side endpoints for retrieval and updates. Furthermore, the software architecture facilitates automated workflows by requiring the reminder module to interact seamlessly with the scheduler component for the execution of timed tasks. The system operates over HTTP or HTTPS depending on the deployment settings, and it supports real-time or near real-time data updates where applicable.

3.2 *Functional Requirements*

3.2.1 User Authentication and Role Management

- The system shall allow new users to create an account using a valid email address.
- The system shall require users to provide complete registration details before account creation.
- The system shall verify user credentials through email confirmation or equivalent verification.
- The system shall allow registered users to log in using valid credentials.
- The system shall authenticate users before granting access to protected features.
- The system shall support the roles Administrator, Organizer, and Participant/Attendee.
- The system shall restrict access to pages and actions based on the user's assigned role.



3.2.2 Event Scheduling

- The system shall allow authorized users to create new events.
- The system shall require complete event details such as title, description, date, time, venue, and expected number of attendees.
- The system shall validate event details before saving the record.
- The system shall allow users to submit event reservation requests for approval.
- The system shall allow users to select a venue and schedule an event based on available time slots.
- The system shall allow administrators or authorized personnel to approve or reject pending reservation requests.
- The system shall update event status as Pending, Approved, or Rejected.
- The system shall notify users about the status of their requests.
- The system shall display event schedules and availability in real-time or near real-time.

3.2.3 Venue Reservation

- The system shall allow users to view the list of available venues.
- The system shall allow users to filter and search venues by capacity, availability, or type.
- The system shall display venue details such as name, location, capacity, and available facilities.
- The system shall provide a visual schedule showing occupied and available time slots.
- The system shall display the availability status of each venue based on the selected date and time.
- The system shall prevent users from selecting venues that are already reserved during the chosen schedule.
- The system shall allow users to reserve a selected venue for a specific date and time.



- The system shall allow users to modify or cancel reservations before approval or within allowed conditions.
- The system shall store and manage reservation records for tracking and reference.

3.2.4 Notification and Reminder System

- The system shall generate automated reminders for upcoming events based on a defined schedule.
- The system shall notify users of changes in event details, including schedule, venue, or time.
- The system shall allow users to configure notification preferences, including timing and notification type.
- The system shall use a scheduling mechanism to trigger time-based reminders automatically.
- The system shall support asynchronous processing of notifications so performance is not heavily affected.
- The system shall allow users to receive, view, and manage notifications through the interface.
- The system shall provide a notification panel where notifications can be marked as read or unread.

3.2.5 Event Dashboard / Overview

- The system shall provide a dashboard that summarizes upcoming events, reservations, and notifications.
- The system shall allow users to view information relevant to their role.
- The system shall display event status, reservation status, and reminder alerts in one centralized view.



- The system shall allow organizers and administrators to monitor event progress efficiently.
- The system shall provide shortcuts to commonly used functions such as event creation, venue reservation, and notification viewing.

3.3 Non-functional Requirements

3.3.1 Security

- User passwords shall be stored securely using hashing or equivalent protection.
- The system shall protect user accounts from unauthorized access.
- Role-based access control shall be enforced for sensitive operations.

3.3.2 Performance

- Login and authentication responses shall be processed within a few seconds under normal conditions.
- Reservation requests shall be processed within an acceptable response time.
- The dashboard shall load quickly even when displaying multiple data items.

3.3.3 Reliability

- The system shall ensure that no double-booking occurs for the same venue and time slot.
- Scheduled reminders shall execute accurately and consistently.
- The system shall minimize the risk of missed notifications.

3.3.4 Usability

- The interface shall be simple, clear, and easy to use on both desktop and mobile devices.
- Users shall complete venue reservation tasks with minimal steps.



- The dashboard shall present information in a clean and organized layout.

3.3.5 Scalability

- The system shall support multiple users accessing and reserving venues simultaneously.
- The system shall handle increasing numbers of events and reservations without major performance loss.

3.4 Logical Structure of the Data

The system stores information in a structured database with the following core entities:

- Users - account information, role, status, and login credentials.
- Events - event title, description, date, time, venue, status, and organizer.
- Venues - venue name, location, capacity, type, facilities, and availability.
- Reservations - request details, approval status, timestamps, and linked event or venue records.
- Notifications - reminder content, recipient, read status, and schedule information.

The main relationships are: one user can create many events; one event is linked to one venue for a specific schedule; one venue can have many reservations across different time slots; and one user can receive many notifications.



4. Supporting Information

4.1 Stakeholders

Stakeholder	Role in the System
Administrators	Approve or reject reservations, manage users, and oversee system activity.
Organizers	Create events, submit requests, and coordinate reservations.
Participants / Attendees	View schedules, receive reminders, and check event details.
Faculty Advisers	Monitor student-led activities and provide oversight.
IT Support / System Maintainers	Maintain the server, database, and application health.